MASTER'S RESEARCH INTERNSHIP (6 MONTHS) • 2027

Instrumental gesture and tipping points in a reed instrument model: effects of two simultaneously time-varying control parameters

This internship is **funded by the French National Research Agency** (ANR) as part of the **BEBOP** project (Beyond Classical Bifurcation Analysis: New Perspectives for Mechanical Engineering).

A **fully funded PhD** on this topic, supported by the same project, may follow the internship.

Background

Many n**onlinear dynamical systems** encountered in physics, biology, climate science, and engineering exhibit multiple stable states for the same set of parameter values—a phenomenon known as **multistability**. **Understanding** and **predicting** the resulting complex dynamics remains a major scientific challenge today.

Over the past few decades, classical bifurcation analysis has become a widely used tool, well beyond research laboratories. While it reveals the possible steady-state regimes (stable and unstable) for a given set of constant parameter values, it does not predict the effect of the time evolution of these parameters. Yet this effect is crucial: in multistable systems with time-varying parameters, the resulting oscillatory regime depends not only on the values reached by the parameters, but also on the **rate at which they change**. In dynamical systems theory, this phenomenon is referred to as **rate-induced tipping** (R-tipping) [1, 2], a concept that has emerged relatively recently, primarily in the context of climate science and ecology.

Self-oscillating musical instruments, such as the saxophone, provide an ideal framework for studying this phenomenon in mechanical engineering. Indeed, (i) they are multistable: the desired note is generally not the only possible stable regime for a given fingering; (ii) during the attack of a note or the transition between notes, the musician varies several control parameters; and (iii) well-known models exist for them, some of which are both simple and representative of complex behaviors, and they are easily amenable to experimentation.

Recent work, carried out in particular at LaMé, has shown that R-tipping sheds light on regime selection under bistability in simplified models of the saxophone [3, 4] and the violin [5]. These results pave the way for a better understanding of the link between instrumental gesture and the sound produced, and highlight the relevance of this phenomenon in mechanical engineering.

Scientific challenges and methodology

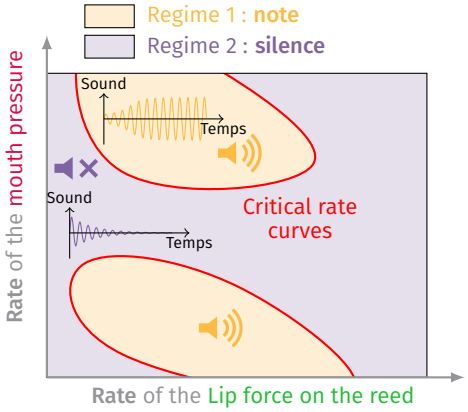
R-tipping theory has mainly focused on **a single time-varying parameter** and has mostly been developed for conceptual climate models. However, in mechanical engineering as in musical acoustics, real systems are driven by **several simultaneously time-varying parameters**. This is typically the case for the saxophonist's gesture, which involves the variation of at least two parameters: the mouth pressure and the lip force on the reed. Predicting rate-induced tipping in such systems remains, to date, a largely open research question.

When a single parameter varies and then settles at a target value, there may exist a **critical rate** beyond which the system tips from one stable state to another. This threshold is a tipping point. A recently introduced tool [3, 4], the **tipping separatrix** (or **R-separatrix**), has made it possible to characterize the tipping point of a simplified bistable saxophone model with a single time-varying parameter (the mouth pressure). For the simplest model [3], the R-separatrix is a particular trajectory in the state space (the amplitude of the mouthpiece pressure), augmented by the mouth pressure treated as an additional variable. It divides this space into two regions: depending on which side the initial condition lies, the system converges to one regime or the other. Since the R-separatrix depends on the rate of change, R-tipping occurs when, for a given initial condition, a change in rate causes the separatrix to move across it. The critical rate is the one for which the initial condition lies exactly on the separatrix.

This internship will allow the student to develop, through pedagogical yet representative examples, **general skills in nonlinear dynamics**, applicable in many fields, from mechanical engineering to climate science, while **addressing a current research question**.

The proposed research approach is as follows:

1. **Selection of a minimal bistable reed instrument model** from existing models, with two control parameters, and computation of its bifurcation diagram in the parameter plane to identify the bistability regions (analytically and using numerical continuation).
2. **Review and reproduction of existing results for a single time-varying parameter** [3, 4]: time-domain simulations, identification of the R-separatrix and determination of the critical tipping rate.
3. **Extension to two simultaneously time-varying parameters**: first mapping of the final regime in the plane of the two rates of change by identifying the **critical rate curves** (see figure).



Profile of the candidate

Master's student (final year) or final-year engineering student in mechanical engineering, acoustics, physics or applied mathematics.

Background in nonlinear dynamical systems and/or nonlinear vibrations, experience with scientific programming, and an interest in mathematical modeling and analysis as well as in numerical simulation and experimental work will be appreciated.

An interest in music or playing a musical instrument would be a plus, but is not required.

Research environment and supervision

The internship will take place in the *Dynamique et Vibrations des Structures* (Structural Dynamics and Vibrations) team (DVS) of the **Laboratoire de Mécanique Gabriel Lamé** (LaMé, UR 7494), on the Blois campus of INSA Centre Val de Loire, in collaboration with the *Laboratoire d'Acoustique de l'Université du Mans* (LAUM, UMR CNRS 6613). The internship will be supervised by:

- Baptiste Bergeot, Associate Professor (HDR), LaMé
- Soizic Terrien, CNRS Researcher, LAUM

Stipend: approximately €680/month (statutory rate).

How to apply

Applications, including a CV, a cover letter, first- and second-year Master's (or equivalent) transcripts and, optionally, contact details of one or two referees, should be sent to:

Baptiste Bergeot (baptiste.bergeot@insa-cvl.fr)

Internship start date: February/March 2027

References

1. Ashwin, P., S. Wieczorek, R. Vitolo, & P. Cox. *Tipping Points in Open Systems: Bifurcation, Noise-Induced and Rate-Dependent Examples in the Climate System*. Philos. Trans. R. Soc. A 370.
2. Wieczorek, S., C. Xie, & P. Ashwin (2023). *Rate-Induced Tipping: Thresholds, Edge States and Connecting Orbits*. Nonlinearity 36, 3238–93.
3. Bergeot, B., S. Terrien, & C. Vergez (2024). Predicting Transient Dynamics in a Model of Reed Musical Instrument with Slowly Time-Varying Control Parameter. Chaos: 34(7) : 073146.
4. Terrien, S., B. Bergeot, C. Vergez, & S. Missoum (2025). *Basins of Attraction in a Dynamical System with a Time-Varying Control Parameter: The Case of Attack Transients in a Simple Model of Reed Musical Instrument*. J. Sound Vib. 618: 119241.